Molecular epidemiology of rabies virus in Vietnam (2006-2009).

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This study was aimed at determining the molecular epidemiology of rabies virus (RABV) circulating in Vietnam. Intra vitam samples (saliva and cerebrospinal fluid) were collected from 31 patients who were believed to have rabies and were admitted to hospitals in northern provinces of Vietnam. Brain samples were collected from 176 sick or furious rabid dogs from all over the country. The human and canine samples were subjected to reverse transcription-polymerase chain reaction analysis. The findings showed that 23 patients tested positive for RABV. Interestingly, 5 rabies patients did not have any history of dog or cat bites, but they had an experience of butchering dogs or cats, or consuming their meat. RABV was also detected in 2 of the 100 sick dogs from slaughterhouses. Molecular epidemiological analysis of 27 RABV strains showed that these viruses could be classified into two groups. The RABVs classified into Group 1 were distributed throughout Vietnam and had sequence similarity with the strains from China, Thailand, Malaysia, and the Philippines. However, the RABVs classified into Group 2 were only found in the northern provinces of Vietnam and showed high sequence similarity with the strain from southern China. This finding suggested the recent influx of Group 2 RABVs between Vietnam and China across the border. Although the incidence of rabies due to circulating RABVs in slaughterhouses is less common than that due to dog bite, the national program for rabies control and prevention in Vietnam should include monitoring of the health of dogs meant for human consumption and vaccination for workers at dog slaughterhouses. Further, monitoring of and research on the circulating RABVs in dog markets may help to determine the cause of rabies and control the spread of rabies in slaughterhouses in Vietnam.